

CLARA Example

From textbook - see textbook for more detailed explanation

Generating data for example

1. Split randomly the data sets in multiple subsets with fixed size (sampsize)

```
set.seed(1234)
# Generate 500 objects, divided into 2 clusters.
df <- rbind(cbind(rnorm(200,0,8), rnorm(200,0,8)),
            cbind(rnorm(300,50,8), rnorm(300,50,8)))

# Specify column and row names
colnames(df) <- c("x", "y")
rownames(df) <- paste0("S", 1:nrow(df))

# Previewing the data
head(df, nrow = 6)

##           x           y
## S1 -9.656526 3.881815
## S2  2.219434 5.574150
## S3  8.675529 1.484111
## S4 -18.765582 5.605868
## S5  3.432998 2.493448
## S6  4.048447 6.083699

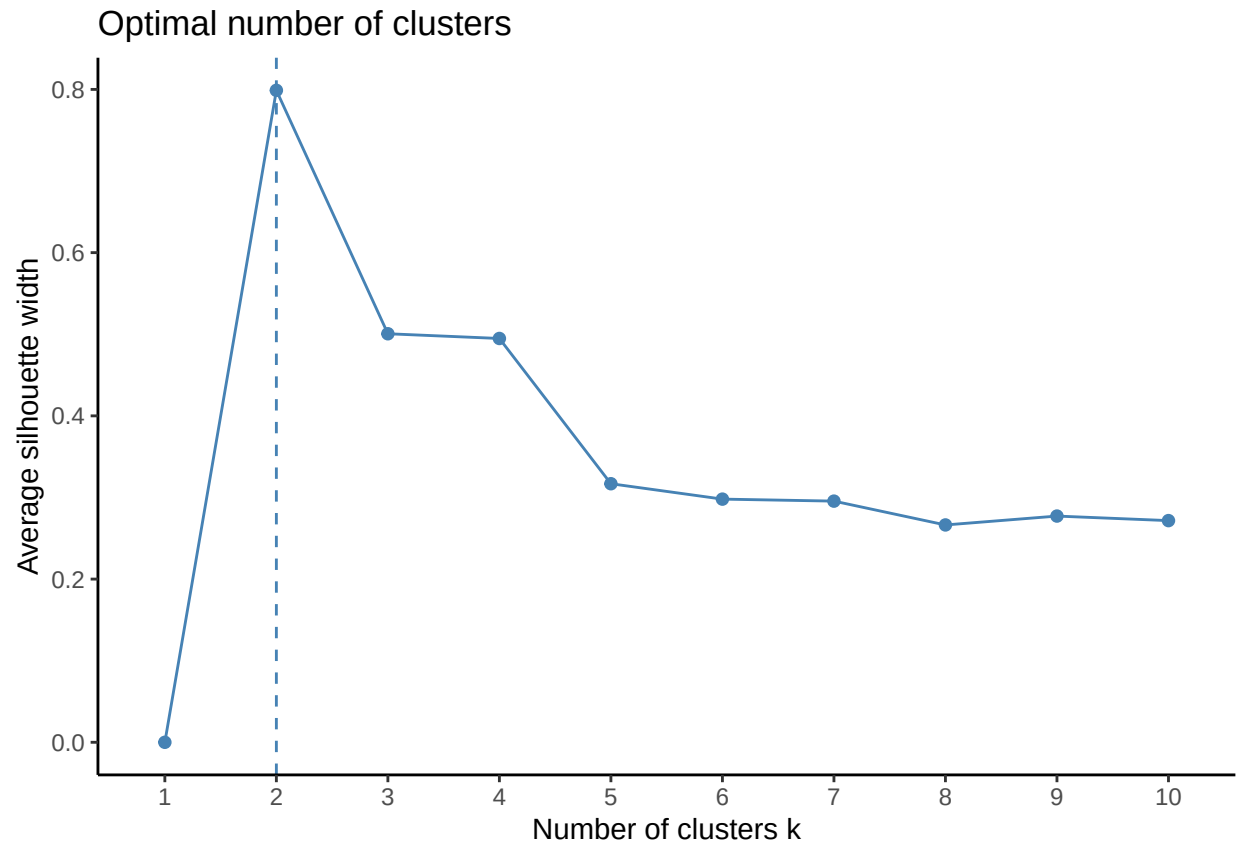
library(cluster)
library(factoextra)

## Warning: package 'factoextra' was built under R version 4.0.5
## Loading required package: ggplot2
## Warning: package 'ggplot2' was built under R version 4.0.5
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa

library(cluster)
library(factoextra)
```

Estimating the optimal number of clusters

```
fviz_nbclust(df, clara, method = "silhouette")+
  theme_classic()
```



From the plot, the suggested number of clusters is 2. In the next section, we'll classify the observations into 2 clusters.

Computing CLARA

```
clara.res <- clara(df, 2, samples = 50, pamLike = TRUE)
```

```
print(clara.res)
```

```
## Call:      clara(x = df, k = 2, samples = 50, pamLike = TRUE)
## Medoids:
##           x           y
## S121 -1.531137  1.145057
## S455 48.357304 50.233499
## Objective function: 9.87862
## Clustering vector:   Named int [1:500] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 ...
## - attr(*, "names")= chr [1:500] "S1" "S2" "S3" "S4" "S5" "S6" "S7" ...
## Cluster sizes:      200 300
## Best sample:
## [1] S37 S49 S54 S63 S68 S71 S76 S80 S82 S101 S103 S108 S109 S118 S121
## [16] S128 S132 S138 S144 S162 S203 S210 S216 S231 S234 S249 S260 S261 S286 S299
## [31] S304 S305 S312 S315 S322 S350 S403 S450 S454 S455 S456 S465 S488 S497
##
## Available components:
## [1] "sample"      "medoids"      "i.med"        "clustering"   "objective"
```

```
## [6] "clusinfo" "diss" "call" "silinfo" "data"
```

If you want to add the point classifications to the original data, use this:

```
dd <- cbind(df, cluster = clara.res$cluster)
head(dd, n = 4)
```

```
##           x           y cluster
## S1 -9.656526 3.881815      1
## S2  2.219434 5.574150      1
## S3  8.675529 1.484111      1
## S4 -18.765582 5.605868      1
```

You can access to the results returned by clara() as follow:

```
# Medoids
clara.res$medoids
```

```
##           x           y
## S121 -1.531137 1.145057
## S455 48.357304 50.233499
```

```
# Clustering
head(clara.res$clustering, 10)
```

```
## S1 S2 S3 S4 S5 S6 S7 S8 S9 S10
##  1  1  1  1  1  1  1  1  1  1
```

Visualizing CLARA clusters

```
fviz_cluster(clara.res,
  palette = c("#00AFBB", "#FC4E07"), # color palette
  ellipse.type = "t", # Concentration ellipse
  geom = "point", pointsize = 1,
  ggtheme = theme_classic()
)
```

